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Jingying Wang

Education

2022 – Present **University of Michigan**, *Ann Arbor, Michigan*
Ph.D. Candidate in Computer Science & Engineering.
Co-advised by Xu Wang (CSE) and Vitaliy Popov (Michigan Medicine)

2020 – 2022 **University of Michigan**, *Ann Arbor, Michigan*
M.S. in Electrical and Computer Engineering.

2016 – 2020 **University of Michigan – Shanghai Jiao Tong University Joint Institute**, *Shanghai, China*
B.S. in Electrical and Computer Engineering.

Professional Experiences

Summer 2026 **Sony Computer Science Laboratories (Sony CSL)**, *Kyoto, Japan*
Research Intern. Host: Jun Rekimoto. [U.5]

Honors, Awards and Fellowships

2026 **UIST 2026 Doctoral Symposium, Detroit, Michigan**

Finalist, CSE Graduate Honors Competition, University of Michigan

2025 **Barbour Scholarship**, University of Michigan (2025–2026)
Rackham's most prestigious scholarship for exceptional women from Asia

2024 **Best Paper Honorable Mention at ACM CHI 2024** [C.6]
Top 5% of submissions

2020 **James Lu and Qingni Li Engineering Graduate Scholarship**, University of Michigan (2020–2021)

Outstanding Graduate, Shanghai Jiao Tong University

2019 **Silver Prize**, UM-SJTU JI Winter Design Expo

2018 **Undergraduate Academic Progress Scholarship** (2018–2019)

Undergraduate Excellent Scholarship (2017–2018, 2018–2019)

Honorable Mention, Mathematical Contest in Modeling (MCM)

Papers Under Review

2026 [U.5] **Jingying Wang**, Son Xuan Nghiem, Xu Wang, Jun Rekimoto. "**CueOnGaze: Leveraging the Attention Differences between Novices and Experts to Deliver Contingent Guidance in Physical Task Training**." Under review.

[U.4] **Jingying Wang**, Rosiana Natalie, Marquise D. Singleterry, Filippos Bellos, Brian George, Gurjit Sandhu, Jason J. Corso, Anhong Guo, Vitaliy Popov, Xu Wang. "**SurgGraph: Quantitative Laparoscopic Video Understanding via Geometry-Grounded Scene Graphs**." Under review. arXiv preprint.

[U.3] **Jingying Wang**, Rosiana Natalie, Keyuan Hu, Wenqian Xu, Brian George, Vitaliy Popov, Anhong Guo, Xu Wang. "**SurgGaze: Implicit Calibration for Accurate Gaze Analysis in Operating Rooms with Wearable Eyetrackers**." Under review. arXiv preprint.

[U.2] Haoting Gao, Kapotaksha Das, Liang Zhang, **Jingying Wang**, Chen Chen, Mohamed Abouelenien, Michael A. Cole, James M. Cooke, Yeyu Wang, Xu Wang, Vitaliy Popov. "**ReadGaze: Turning Gaze Traces into Inspectable Evidence for Clinical Simulation Debriefing**." Under review.

[U.1] Yayuan Li, Chenglin Li, **Jingying Wang**, Filippos Bellos, Anhong Guo, Jason J. Corso. "**Substantial, Decomposable, and Invisible: Visual Context Misalignment in Instructional Videos for Physical Tasks**." Under review.

Peer-Reviewed Full Papers

2025 [C.8] **Jingying Wang**, Jingjing Zhang, Juana Nicoll Capizzano, Matthew Sigakis, Xu Wang*, Vitaliy Popov*. "**eXplainMR: Generating Real-time Textual and Visual eXplanations to Facilitate UltraSonography Learning in MR**." In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI'25). Yokohama, Japan. 2025.

2024 [C.7] **Jingying Wang**, Haoran Tang, Taylor Kantor, Tandis Soltani, Vitaliy Popov, Xu Wang. "**Surgment: Segmentation-enabled Semantic Search and Creation of Visual Question and Feedback to Support Video-Based Surgery Learning**." In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (CHI'24). Honolulu, Hawaii. 2024.

[C.6] Xinyue Chen*, Vitaliy Popov*, **Jingying Wang**, Michael Kemp, Gurjit Sandhu, Taylor Kantor, Natalie Mateju, Xu Wang. "**Looking Together $\neq$ Seeing the Same Thing: Understanding Surgeons' Visual Needs During Intra-operative Coordination and Instruction**." In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (CHI'24). Honolulu, Hawaii. 2024. *Best Paper Honorable Mention*

2023 [C.5] Ye Pan, Ruisi Zhang, **Jingying Wang**, Yu Ding, Kenny Mitchell. "**Real-time Facial Animation for 3D Stylized Character with Emotion Dynamics**." In *Proceedings of the 31st ACM International Conference on Multimedia* (MM'23). Ottawa, Canada. 2023.

[C.4] **Jingying Wang**, Yilin Qiu, Keyu Chen, Yu Ding, Ye Pan. "**Fully Automatic Blendshape Generation for Stylized Characters**." In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces* (IEEE VR'23). Shanghai, China. 2023.

[C.3] Yukang Yan, Haohua Liu, Yingtian Shi, **Jingying Wang**, Ruici Guo, Zisu Li, Xuhai Xu, Chun Yu, Yuntao Wang, Yuanchun Shi. "**ConeSpeech: Exploring Directional Speech Interaction for Multi-Person Remote Communication in Virtual Reality**." *IEEE Transactions on Visualization and Computer Graphics* (TVCG, IEEE VR'23). 2023.

2021 [C.2] **Jingying Wang**. "**A Survey on Crowd Counting Methods and Datasets**." In *Advances in Computer, Communication and Computational Sciences*, Springer. 2021.

2020 [C.1] Tianyu Wang, Minhao Jin, **Jingying Wang**, Yijie Wang, Mian Li. "**Towards a Data-Driven Method for RGB Video-Based Hand Action Quality Assessment in Real Time**." In *Proceedings of the 35th Annual ACM Symposium on Applied Computing* (SAC'20). Brno, Czech Republic. 2020.

Peer-Reviewed Workshops, Posters, and Demos

2023 [A.2] **Jingying Wang**, Vitaliy Popov, Xu Wang. "**SketchSearch: Fine-tuning Reference Maps to Create Exercises in Support of Video-based Learning for Surgeons**." In *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (UIST'23 Demo). San Francisco, California. 2023.

2022 [A.1] Ye Pan, Ruisi Zhang, **Jingying Wang**, Nengfu Chen, Yilin Qiu, Yu Ding, Kenny Mitchell. "**MienCap: Performance-based Facial Animation with Live Mood Dynamics**." In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops* (IEEE VRW'22). 2022.

Research Experiences Before Ph.D.

2021 – 2022 **Information Interaction Lab (Mi2 Lab), University of Michigan**
Research Assistant. Advisors: Michael Nebeling and Yukang Yan.

2020 – 2023 **Character Lab, John Hopcroft Center, Shanghai Jiao Tong University**
Research Assistant. Advisor: Ye Pan. [C.4, C.5, A.1]

2018 – 2020 **System, Control, and Optimization Laboratory, UM-SJTU Joint Institute**
Undergraduate Research Assistant. Advisor: Mian Li. [C.1]

Academic Services

Reviewing

Conferences
CHI ('25, '26, '27), UIST ('25, '26), ISMAR ('25, '26), IEEE VR ('23),
CHI Late-Breaking Work ('23, '24, '25), CSCW Posters ('23)

Student Volunteer

CHI'24

Invited Talks and Presentations

Nov 2026 Surgical Practice AI Research Community (SPARC), Department of Surgery & Center for Healthcare Outcomes and Policy, University of Michigan. (*upcoming*)

Nov 2026 CSE Graduate Honors Competition, University of Michigan. (*upcoming*)

Nov 2026 UIST 2026 Doctoral Symposium, Detroit, Michigan. (*upcoming*)

Oct 2026 DCM&B Tools and Technology Seminar, University of Michigan. "Human-AI Systems for Externalizing Surgical Expertise."

Apr 2026 Guest Lecturer, EECS 598 (Instructor: Anhong Guo), University of Michigan. "Human-AI Systems for Enhancing Visual Understanding in Surgical Training."

Feb 2026 VIGIL All Hands Meeting, Michigan Robotics, University of Michigan. "Gaze Calibration in the Operating Room."

Jun 2025 VIGIL All Hands Meeting, Michigan Robotics, University of Michigan. "UltraSonography Learning in MR."

Dec 2024 CSTAR Research in Progress Session, Center for Surgical Training and Research, University of Michigan.

Nov 2023 DLHS / HILS Research Seminar, University of Michigan Medical School. "What to Do About and With AI to Enhance Medical Education and Clinical Simulation: Use Cases from Surgery and Emergency Medicine." (Joint talk with Vitaliy Popov)

Teaching Experiences

2021 EECS 551 Matrix Methods for Signal Processing, Data Analysis and Machine Learning, University of Michigan. Grader.

Winter 2024 EECS 496 Major Design Experience – Professionalism, University of Michigan. Graduate Student Instructor.

2019 EECS 445 Introduction to Machine Learning, UM-SJTU Joint Institute. Teaching Assistant.

Technical Skills

Programming Python, C#, C++, Julia, JavaScript, MATLAB, C, Verilog HDL

Frameworks PyTorch, TensorFlow, OpenCV, Unity3D, ReactJS, OpenGL, Maya

Hardware Tobii & Pupil Labs eye trackers, Meta Quest, HTC VIVE, PICO, Leap Motion, Arduino, FPGA